CAPTURING AND ANALYSING PACKETS USING WIRESHARK

AIM:

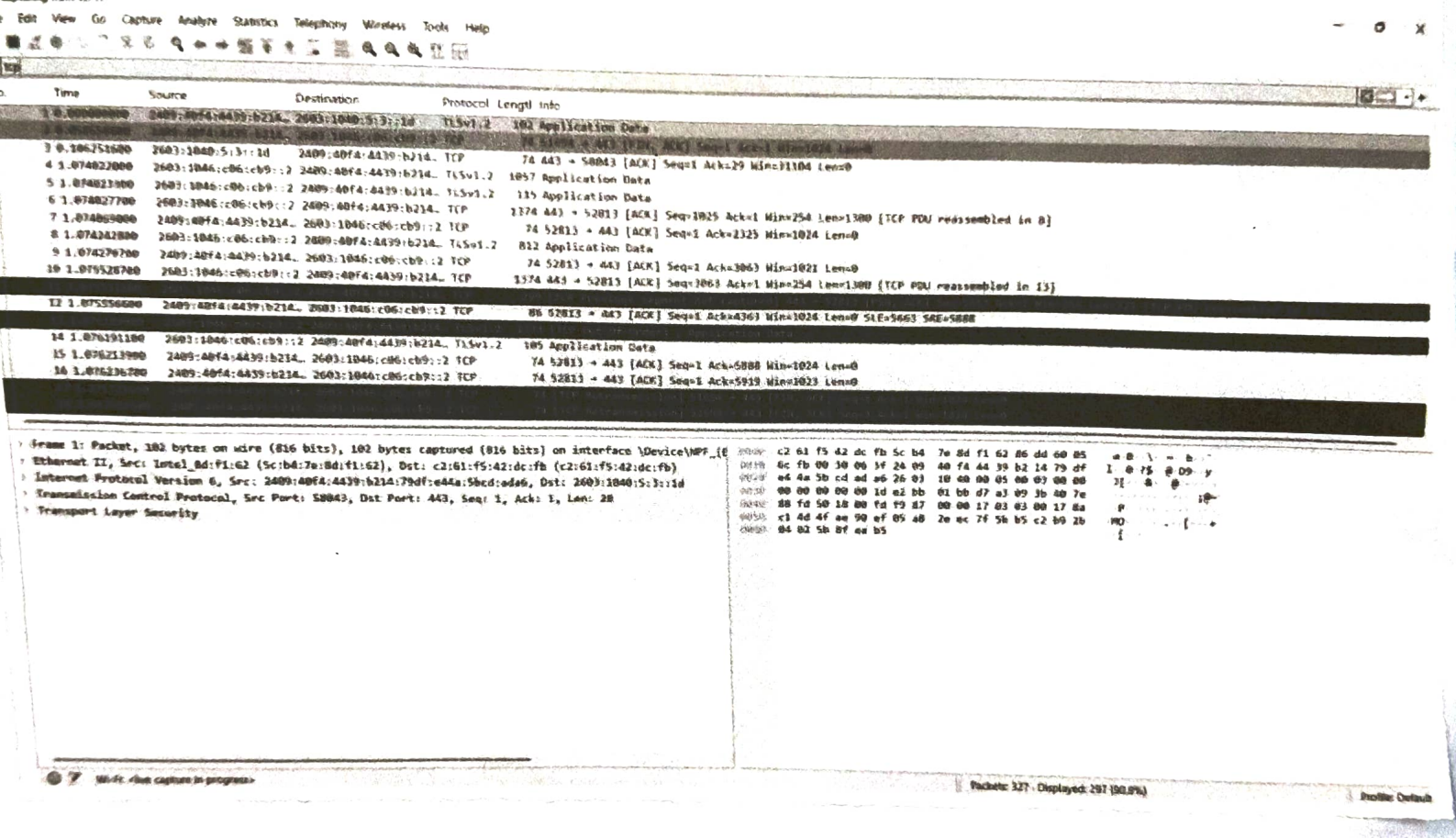
To capture, filter, inspect and analyse network packets using the wireshark network protocol analyzer.

WIRESHARK:

wireshark is an open-source network protocol analyzer that captures packets in real time and displays them in a human-readable formed. It helps network administrators, developers and analyze different protocol using filters & packets.

ALGORITHM:

- Install and launch wireshark
- Select the active network interface (Ethernet) after
- 100 packets.
- Set capture → Stop capture automatically after 100 packets.
- click start capture
- Capture the required packets
- Apply display filters (TCP / UDP, ARP, DNS, HTTPP))
- Inspect packet details and bytes
- View the Flow Graph
- Save the captured packets.


Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
No. Time Source Destination Protocol Length Info
3 0.106751600 2603:1040:5:3::1d 2409:40f4:4439:b214... TCP 74 443 → 58043 [ACK] Seq=1 Ack=29 Win=1104 Len=0
4 1.074022000 2603:1046:c06:cb9::2 2409:40f4:4439:b214... TLSv1.2 1057 Application Data
5 1.074023000 2603:1046:c06:cb9::2 2409:40f4:4439:b214... TLSv1.2 115 Application Data
6 1.074027700 2603:1046:c06:cb9::2 2409:40f4:4439:b214... TCP 1374 443 → 52813 [ACK] Seq=1025 Ack=1 Win=254 Len=1300 [TCP PDU reassembled in 8]
7 1.074059000 2409:40f4:4439:b214... 2603:1046:c06:cb9::2 TCP 74 52813 → 443 [ACK] Seq=1 Ack=2325 Win=1024 Len=0
8 1.074342800 2603:1046:c06:cb9::2 2409:40f4:4439:b214... TLSv1.2 812 Application Data
9 1.074270700 2409:40f4:4439:b214... 2603:1046:c06:cb9::2 TCP 74 52813 → 443 [ACK] Seq=1 Ack=3063 Win=1021 Len=0
10 1.075528700 2603:1046:c06:cb9::2 2409:40f4:4439:b214... TCP 1374 443 → 52813 [ACK] Seq=3063 Ack=1 Win=254 Len=1300 [TCP PDU reassembled in 13]
12 1.075556600 2409:40f4:4439:b214... 2603:1046:c06:cb9::2 TCP 86 52813 → 443 [ACK] Seq=1 Ack=4363 Win=1024 Len=0 SLE=5663 SRE=5888
14 1.076191100 2603:1046:c06:cb9::2 2409:40f4:4439:b214... TLSv1.2 105 Application Data
15 1.076213900 2409:40f4:4439:b214... 2603:1046:c06:cb9::2 TCP 74 52813 → 443 [ACK] Seq=1 Ack=5888 Win=1024 Len=0
16 1.076236700 2409:40f4:4439:b214... 2603:1046:c06:cb9::2 TCP 74 52813 → 443 [ACK] Seq=1 Ack=5919 Win=1023 Len=0
Frame 1: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface \Device\NPF_{
Ethernet II, Src: Intel_8d:f1:62 (5c:b4:7e:8d:f1:62), Dst: c2:61:f5:42:dc:fb (c2:61:f5:42:dc:fb)
Internet Protocol Version 6, Src: 2409:40f4:4439:b214:79df:e44a:5bcd:eda6, Dst: 2603:1040:5:3::1d
Transmission Control Protocol, Src Port: 58043, Dst Port: 443, Seq: 1, Ack: 1, Len: 28
Transport Layer Security
c2 61 f5 42 dc fb 5c b4 7e 8d f1 62 86 dd 60 05
6c fb 00 30 06 3f 24 09 40 f4 44 39 b2 14 79 df
e4 4a 5b cd ed a6 26 03 10 40 00 05 00 09 00 00
00 00 00 00 00 1d e2 bb 01 bb d7 a3 09 3b 4b 7e
18 fd 50 18 80 fd f9 87 00 00 17 03 03 00 17 8a
c1 4d 4f ee 90 ef 05 48 2e ec 7f 5b b5 c2 b9 2b
04 82 5b 8f ee b5
Wi-Fi: <live capture in progress>
Packets: 327 · Displayed: 297 (90.8%)
Profile: Default

FEATURES OF WIRESHARK:

- Captures live network - traffic
- Supports protocol analysis
- Provide capture and display filters

MAIN - COMPONENTS

1). Packet list Pant:

Displays all captured packets with information such as sources, destination, protocol, length and time

2). Packet Details Pane:

Displays detailed protocols headers and fields of the selected packet.

3) Packet Bytes Pane:

Displays the raw packet data in hexadecimal and ASCII format.

FILTERS:

- tcp || udp

procedure:

- Start packet capture
- Apply the filter
- Inspect TCP / UDP packets
- View statistics → Flow Graph
- Save the capture

- arp

procedure

- Start packet capture